

Appendix B

Short time asymptotics

The asymptotic treatment of diffusion is based upon the idea that, for sufficiently short times, the effects of the interface are only significant at locations sufficiently close to it that the interface can be approximated by an infinite flat plane. This idea was apparently first expressed in Ref. [166] and was implemented in detail in Ref. [167], where some of the results described in this appendix were first obtained. In that case, the problem reduces to diffusion in a semi-infinite space bounded by that plane, which we take to be $x = 0$.

The diffusion propagator for that problem is a product of three one-dimensional diffusion propagators—free or unrestricted propagators G_0 along y and z , e.g.

$$G_0(y, y'|t) = \left(\frac{1}{4\pi D_p t} \right)^{1/2} e^{-\frac{(y-y')^2}{4D_p t}},$$

and one “half space propagator” $G_1(x, x'|t)$ along x . The latter can be written in closed form if $\rho = 0$, and its Laplace transform $\tilde{G}_1(x, x'|s) \equiv \mathcal{L}[G_1(x, x'|t)]$ can be written in closed form when ρ is arbitrary:

$$\begin{aligned} \tilde{G}_1(x, x'|s) &\equiv \mathcal{L}[G_1(x, x'|t)] \equiv \int_0^\infty dt G_1(x, x'|t) e^{-st} \\ &= \frac{1}{2(sD_p)^{1/2}} \left[e^{-(s/D_p)^{1/2}|x-x'|} + \frac{(sD_p)^{1/2} - \rho}{(sD_p)^{1/2} + \rho} e^{-(s/D_p)^{1/2}(x+x')} \right]. \end{aligned}$$

The Laplace transform of the relative total polarization $M(t)/M(0) = M(\mathbf{k} = 0, t)$ is then given by

$$\mathcal{L}[M(\mathbf{k} = 0, t)] = \frac{1}{s} - \frac{S_p}{V_p} \frac{D_p^{1/2} \rho}{s^{3/2} [(sD_p)^{1/2} + \rho]}, \quad (\text{B.1})$$

and the transform of the product $M(\mathbf{k} = 0, t)D(t)t$, where $D(t)$ is defined by Eq. (7.37) or Eq. (A.16), is given by

$$\begin{aligned} \mathcal{L}[M(\mathbf{k} = 0, t)D(t)t] = & \frac{D_p}{s^2} - \frac{S_p}{V_p} \frac{D_p}{s^2} \left[\frac{2D_p\rho}{(sD_p)^{1/2}[(sD_p)^{1/2} + \rho]} + \right. \\ & \left. + \frac{D_p}{3(sD_p)^{1/2}} \frac{(sD_p)^{1/2} - \rho}{(sD_p)^{1/2} + \rho} + \frac{D_p\rho}{3[(sD_p)^{1/2} + \rho]^2} \right]. \quad (\text{B.2}) \end{aligned}$$

These results are valid as long as $(D_p/s)^{1/2}$ is much less than any finite size feature of the interface, and they can be used to reconstruct $M(t)$, $D(t)$ for times such that $(D_p t)^{1/2}$ is similarly small. This is most conveniently done, in the two limits $\rho t \ll (D_p t)^{1/2}$ and $\rho t \gg (D_p t)^{1/2}$, by first expanding Eqs. (B.1), (B.2) in powers of ρ or $1/\rho$ and then applying the inverse Laplace transformation term by term. This leads to the following results [18]:

$$\frac{M(t)}{M(0)} \cong \begin{cases} 1 - \frac{S_p}{V_p} \rho t & \text{for } \rho \ll (D_p/t)^{1/2}, \\ 1 - 2 \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} & \text{for } \rho \gg (D_p/t)^{1/2}, \end{cases} \quad (\text{B.3})$$

$$\frac{D(t)}{D_p} \cong \begin{cases} 1 - \frac{4}{9} \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} + \frac{1}{6} \frac{S_p}{V_p} \rho t & \text{for } \rho \ll (D_p/t)^{1/2}, \\ 1 - \frac{2}{9} \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} & \text{for } \rho \gg (D_p/t)^{1/2}. \end{cases} \quad (\text{B.4})$$

If we use the more physical definition of $D(t)$, namely Eqs. (A.14) and (A.15), we get some slightly different results [15]:

$$\frac{D(t)}{D_p} \cong \begin{cases} 1 - \frac{4}{9} \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} + \frac{3}{4} \frac{S_p}{V_p} \rho t & \text{for } \rho \ll (D_p/t)^{1/2}, \\ 1 - 0.392 \frac{2}{9} \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} & \text{for } \rho \gg (D_p/t)^{1/2}. \end{cases} \quad (\text{B.5})$$

The result in the second line is based on the following integral ($\text{erfc}(x)$ is the complementary error function—see Ref. [2])

$$6 \int_0^\infty \frac{2\sqrt{\pi}x^2 \text{erfc}(x) - xe^{-x^2}}{1 - \text{erfc}(x)} dx \cong 6 \cdot 0.065352 = 0.392112,$$

which was evaluated numerically using Mathematica 4. Interestingly, the leading order short time correction to D_p is independent of ρ according to both definitions of $D(t)$. At the shortest times, that leading correction is the same for both definitions of $D(t)$ —compare the first line in Eqs. (B.4) and (B.5).

Similar considerations, applied to Eq. (7.70), lead to the short time result for $P_s(t)$ of Eq. (7.71)—see Ref. [110].

All the results obtained in this appendix do not include corrections that arise from microgeometrical details other than S_p/V_p , such as the local radii of curvature of the interface. Some of those corrections, which are of higher order in some appropriate characteristic size of the pore space, are discussed and calculated in Ref. [167], while others appear in Appendix C of Ref. [18].